

Practice of Epidemiology

Machine-Learning–Based Forecasting of Dengue Fever in Brazilian Cities Using Epidemiologic and Meteorological Variables

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Initially submitted April 1, 2021; accepted for publication May 10, 2022.

Dengue is a serious public health concern in Brazil and globally. In the absence of a universal vaccine or specific treatments, prevention relies on vector control and disease surveillance. Accurate and early forecasts can help reduce the spread of the disease. In this study, we developed a model for predicting monthly dengue cases in Brazilian cities 1 month ahead, using data from 2007–2019. We compared different machine learning algorithms and feature selection methods using epidemiologic and meteorological variables. We found that different models worked best in different cities, and a random forests model trained on monthly dengue cases performed best overall. It produced lower errors than a seasonal naive baseline model, gradient boosting regression, a feed-forward neural network, or support vector regression. For each city, we computed the mean absolute error between predictions and true monthly numbers of dengue cases on the test data set. The median error across all cities was 12.2 cases. This error was reduced to 11.9 when selecting the optimal combination of algorithm and input features for each city individually. Machine learning and especially decision tree ensemble models may contribute to dengue surveillance in Brazil, as they produce low out-of-sample prediction errors for a geographically diverse set of cities.

dengue; epidemiologic methods; feature selection; machine learning; prediction

Abbreviations: MAE, mean absolute error; PC, Peter-Clark; PCMC, Peter-Clark momentary conditional independence; RMSE, root mean squared error; SVR, support vector regression.

Dengue fever is a serious public health concern, burdening individuals' lives and national economies. It is a mosquito-borne infectious disease that affects an estimated 105 million people each year (1). Half of the world's population and 129 countries are at risk of infection (2).

Brazil is home to about half of all reported dengue cases in the Americas (3). In the 3 decades after 1986, when official national reporting began, over 11 million suspected cases and over 5,000 confirmed deaths were reported. Both infection and fatality rates increased during this period. The southeastern and northeastern regions of Brazil are particularly affected, but nearly all Brazilian states experience dengue-related deaths. Several outbreaks have occurred, primarily due to reintroduction of one of the 4 dengue virus serotypes. In 2007, the serotype DENV-2 reemerged, causing an epidemic that disproportionately affected children, who accounted for over half of the epidemic's deaths (4).

Besides its human cost, dengue has an important impact on the Brazilian economy. During the 2013 outbreak alone, the economic burden of workforce absenteeism due to dengue was estimated at \$300 million (5).

There is no universal vaccine against dengue, and few treatment options exist. Prevention relies on methods such as vector control, which underscores the importance of disease surveillance (6). To best inform public health decisions such as resource allocation, disease forecasts need to be available early and at a granular geospatial resolution, while maintaining a high level of accuracy. Yet dengue fever is difficult to predict. Firstly, infection with one of the 4 dengue serotypes provides lifetime immunity to that serotype, as well as temporary cross-immunity to other serotypes. Secondly, dengue incidence is influenced by a wide range of factors, including climate conditions, human mobility (7), land use (8), and socioeconomic status (9). The relationship

between dengue and climate in particular has been extensively studied, and associations have been found with rainfall (10, 11), climate change (12, 13), temperature (10, 11, 14, 15), extreme weather events such as El Niño and La Niña (10, 15), humidity (14), atmospheric pressure (15), and sea surface temperatures (11, 16). These factors may have non-linear, context-specific, and time-variant effects on disease incidence. This broad range of risk factors, together with temporary cross-immunity between serotypes, results in an irregular periodicity of outbreaks, which poses a challenge to disease modeling.

The existing literature on dengue covers a range of forecasting approaches, including both theoretical and data-centric methods. The classical epidemiologic approach is compartmental modeling, which assesses the evolution of the number of infectious individuals and other compartments within a population (17, 18). Agent-based models simulate the actions of individuals and can provide geospatial estimates of the spread of disease. They are helpful in assessing the potential impacts of policies that alter individual behaviors or environmental conditions, such as the release of sterile mosquitoes (19, 20). Statistical time-series forecasting tools, such as the seasonal autoregressive integrated moving average (SARIMA) model, employ a data-centric approach and leverage the highly autocorrelated nature of dengue to predict future incidence (21, 22).

Machine learning models take advantage of the increasing measurement capacity and public availability of epidemiologic and other data sources, and they often incorporate novel “big data” streams, such as social media activity, mobile phone data, or search engine queries. Relative to other forecasting approaches, they rely on few assumptions, are nonparametric, and operate well in high-dimensional spaces with many covariates. Rather than defining a model based on what is known about the disease, machine learning models estimate their functional form and parameters directly from the data. At the expense of reduced explainability, approaches such as neural networks have demonstrated a powerful forecasting ability for diseases like malaria (23), influenza (24), and coronavirus disease 2019 (25).

Machine learning models have also been applied to forecast dengue in different contexts, often incorporating climate data. Baquero et al. (26), for example, compared the performance of machine learning and statistical models to forecast the number of dengue cases in the city of São Paulo, Brazil, utilizing measurements of temperature, rainfall, and humidity. In a study by Andersson et al. (27), dengue risk was assessed even more granularly in Rio de Janeiro, Brazil. The authors used convolutional neural networks based on aerial and street-view images to predict dengue risk at the neighborhood level. The literature shows that the most effective machine learning model for dengue prediction varies across contexts, such as data inputs or location. Guo et al. (28) used epidemiologic, climate, and Baidu search data (Baidu, Inc., Beijing, China) to forecast the number of dengue cases in a Chinese province. They compared several different algorithms and achieved the best results with a support vector regression (SVR) model. Kesorn et al. (29) also achieved the best performance with an SVR model for

classifying the risk of dengue hemorrhagic fever in Thailand given data on climate, population density, and mosquito infection rates. Xu et al. (30) developed a recurrent neural network model with a long short-term memory layer to predict dengue in 20 Chinese cities and also included climate variables. Their model outperformed an SVR model.

In this study, we assessed the potential of machine learning models to predict dengue 1 month ahead in over 200 Brazilian cities. We compared different algorithms, including decision tree ensemble approaches, neural networks, SVR, and a seasonal naive baseline model. We also compared different sets of meteorological and epidemiologic variables to better understand their predictive contribution. The best model for each city was selected using expanding time-series cross-validation and was tested on a holdout test data set. The best performance was achieved by the random forests algorithm. While climate variables improved predictions in some cities, the most important predictors across all cities were numbers of dengue cases in prior months. Our model outperformed existing comparable approaches by achieving a lower forecast error for São Paulo than the model of Baquero et al. (26).

METHODS

Data

We used official government sources for data on epidemiologic and meteorological variables. Monthly dengue cases were reported for each municipality in Brazil's Notifiable Diseases Information System (31) for the years spanning 2007–2019, counting cases of dengue fever with and without warning signs as well as severe dengue. In the interest of developing a model with practical public health value, we used data on suspected cases, since this information was available early, rather than relying only on confirmed case data, which were reported with delay. The National Meteorological Institute provided daily meteorological measurements (32), which we collected for the period from January 1, 2005, to December 31, 2019, and aggregated to monthly time series using both the mean and the standard deviation. This helped to account for the overall quantity as well as the variability in climate. The included variables were 1) rainfall, 2) maximum temperature, 3) minimum temperature, 4) relative median temperature, 5) insolation (the amount of solar energy reaching the earth), 6) rate of evaporation, 7) median relative humidity, and 8) median wind speed.

We merged the 2 data sources using the coordinates of the weather stations and the geographic boundaries of municipalities. The combined data set covered 234 municipalities. Data were split into a training set (9 years; January 2007–December 2015) and a holdout test set (4 years; January 2016–December 2019). The data were normalized to have a mean of 0 and a standard deviation of 1. The normalization was performed using the training data only. During cross-validation, the mean and standard deviation were computed separately on each training fold. A relatively large test set was chosen to account for the wide range of epidemiologic scenarios occurring in previous years: the Zika epidemic in

2016, followed by exceptionally low dengue incidence in 2017 and 2018, and finally large outbreaks in 2019.

Feature selection

We compared 4 different sets of independent variables, 2 of which were selected naively and 2 of which were selected using relationship metrics.

- The first set contained the past 11 months of dengue cases.
- The second set of features also included the past 11 months of all climate variables, in addition to 11 months of dengue cases. Because we used 8 different climate variables, aggregated to monthly time series using 2 statistics (mean and standard deviation) and 11 months of each (see data description above), we had a total of 187 epidemiologic and meteorological features.
- To reduce the number of covariates, the third set of independent variables was determined by means of the Peter-Clark momentary conditional independence (PCMCI) causal discovery algorithm. Seven variables were selected at significance level $\alpha = 0.05$. The selected variables were dengue (months 1–5 and 11) and mean rainfall (month 4).
- The fourth set was made up of the variables that were most strongly correlated with the number of dengue cases, as measured by the Pearson correlation coefficient. The number of variables was fixed to 7, so as to allow for direct comparison with the PCMCI approach. The selected variables were dengue months 1–4 and 9–11.

Causal feature selection is increasingly being explored as an alternative to correlation-based selection or a-priori model specification, as it offers the potential for greater interpretability and more robust models (33–36).

In this analysis, we employed the PCMCI algorithm (37), a constraint-based causal discovery method, which aims to identify which variables in a data set are causally related. Constraint-based algorithms leverage the dependencies between variables to reconstruct the causal graph underlying the data generation mechanism. A classic constraint-based approach is the Peter-Clark (PC) algorithm, named after Peter Spirtes and Clark Glymour (38). It is based on the following idea: If 2 variables are independent of each other, they cannot constitute direct cause and effect, meaning they cannot be linked in a causal graph. Similarly, if 2 variables are independent, conditioning on a set of other variables, they are also not directly linked. Repeatedly testing for such conditional and unconditional independencies helps us construct the skeleton of a causal graph, where edges represent possible causal links. To differentiate cause from effect, the PC algorithm then assigns directionality to the edges of the graph according to a set of orientation propagation rules (see Glymour et al. (39) for an overview).

A challenge in causal discovery is achieving a high discovery rate of true causal links while avoiding false-positive links. In the PC algorithm, this trade-off is determined by the significance level of the independence test. The PCMCI algorithm aims to resolve this challenge by implementing 2

stages of independence tests, which also give the algorithm its name: The first stage consists of a modified version of the PC algorithm with a generous significance level, α_{PC} , to ensure a high rate of discovery of true links; the second stage aims to remove false-positive links via a momentary conditional independence test (see Runge et al. (37) for details). PCMCI was developed for time-series analysis and therefore considers as nodes in the network up to τ past observations of all the variables. Different kinds of independence tests can be performed in the first stage, including the partial correlation test.

Prediction

We compared the following forecasting algorithms. As a baseline, we implemented a seasonal naive model (40), which predicts the number of cases y in a given month t to be equal to the number of cases that occurred in the same month of the previous year: $\hat{y}_t = y_{t-12}$. We compared the baseline model with the following machine learning algorithms.

Random forests (41) uses an ensemble of decision trees to make predictions. Each tree is grown by selecting a subset of observations in the training set with replacement (bootstrap aggregating, i.e., bagging) and then determining their best split based on a random subset of features using the mean squared error as a splitting criterion.

Another tree-based ensemble approach is gradient boosting regression (42). As with random forests, final predictions are determined by the combined results of several decision trees. Unlike random forests, gradient boosting regression builds trees one at a time, applying boosting at each iteration. Boosting is the process of giving a higher weight to examples that are difficult to predict, thus incentivizing the model to improve its forecasts for the examples it predicted incorrectly previously.

SVR (43, 44) is a statistical learning technique that first transforms the input space using a kernel function and then fits a linear function on the data. We used the radial basis function as the kernel. SVR is ϵ -insensitive, meaning errors of absolute magnitude up to ϵ are ignored, but errors that fall outside this range are minimized, while at the same time maintaining flatness of the fitted function (see Smola and Schölkopf (45) and Cristianini and Shawe-Taylor (46) for details).

Finally, we implemented a multilayer perceptron neural network, comparing different architectures of up to 3 layers with up to 128 hidden units. To produce predictions, each neuron (a processing unit) in the network combines and transforms the input variables before passing them on to the next layer of neurons. Transformations entail weighting and linearly combining the independent variables together with a bias term, and then applying a nonlinear function (activation). The weights and biases involved in these transformations are optimized according to a loss function (error measure) using gradient descent. This learning process aims to minimize the prediction errors (for more details, see Haykin (47)). We used rectified linear unit (ReLU) activation, Adam optimization, 500 epochs, a learning rate of 0.001, a mean squared error loss function, and a mini-batch size of 200 observations. To avoid overfitting our predictions

Table 1. Machine Learning Hyperparameter Ranges Considered for Dengue Prediction Models, Brazil, 2007–2015

Algorithm and Hyperparameter	Values Tested
Random forests	
No. of trees	50, 100, 150, 200
Maximum tree depth	2, 3, 6, none
Gradient boosting regression	
No. of trees	50, 100, 150, 200
Maximum tree depth	2, 3, 6, none
Multilayer perceptron	
No. of layers	1–3
No. of units per layer	32, 64, 128
Support vector regression	
ε	0.01, 0.05, 0.1, 0.2

to the training examples, we implemented ridge regression, adding an L2 regularization term to the cost with a parameter of 0.2.

We tested several hyperparameters for each of the machine learning algorithms, which are presented in Table 1. The optimal hyperparameters are selected during the validation stage, and their values vary across the sets of independent variables. The chosen optimal values are presented in Table 2.

Model evaluation

We used the root mean squared error (RMSE) (equation 1) and the mean absolute error (MAE) (equation 2) for model evaluation and selection:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (1)$$

and

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|, \quad (2)$$

where N is the total number of observations, $\hat{y}$ are the predicted values, and y are the actually observed numbers of monthly cases.

Models were selected using expanding time-series cross-validation. Traditional cross-validation would involve shuffling observations, which is not possible with sequence data. Instead, we maintained the chronological order of the data, by shifting our validation fold for each iteration of cross-validation. We began with 1 year of training data (year 2007) and the following year of validation data (year 2008). We then shifted the validation fold to the following year (2009) and used all previous observations for training (years

Table 2. Optimal Hyperparameters Selected Using Cross-Validation for Each Set of Input Variables in Dengue Prediction Models, Brazil, 2007–2015

Algorithm and Feature Set	Hyperparameter Value
Random forests	
Only dengue, correlation	
No. of trees	50
Maximum tree depth	None
With climate	
No. of trees	200
Maximum tree depth	6
PCMCi	
No. of trees	100
Maximum tree depth	6
Gradient boosting regression	
All	
No. of trees	200
Maximum tree depth	2
Multilayer perceptron	
Only dengue	
Architecture	32, 32
With climate, correlation	
Architecture	32
PCMCi	
Architecture	64
Support vector regression	
Only dengue, PCMCi, correlation	
ε	0.05
With climate	
ε	0.01

Abbreviation: PCMCi, Peter-Clark momentary conditional independence.

2007–2008). This shifting process was repeated until the full training data were covered, resulting in 8 separate validation folds. The validation error of each city was computed as an average across the 8 folds.

RESULTS

The random forests algorithm performed best on the validation folds, especially with only dengue variables. This combination of algorithm and feature set had the lowest errors across all cities according to the mean MAE and RMSE. For the median city, this resulted in an RMSE of 15.5 and predictions on average within 8.6 cases (Table 3). PCMCi feature selection gave the lowest median MAE in combination with the random forests algorithm. The lowest median RMSE was achieved by the gradient boosting regression model trained on only dengue variables. When selecting

Table 3. Dengue Prediction Errors on the Validation Data Set, Brazil, 2007–2015

Algorithm and Feature Set	MAE		RMSE	
	Mean	Median	Mean	Median
Gradient boosting regression				
Correlation	39.0	8.8	68.1	13.6
PCMCi	38.7	8.9	67.6	13.9
Only dengue ^a	38.3	8.9	67.3	13.5
With climate	41.4	9.7	72.5	15.6
Multilayer perceptron				
Correlation	44.0	12.1	74.6	17.7
PCMCi	41.7	12.4	71.2	18.7
Only dengue	48.2	13.0	78.6	18.3
With climate	53.8	22.7	89.9	32.0
Random forests				
Correlation	37.4	8.9	67.4	15.4
PCMCi ^b	38.8	8.5	67.9	14.2
Only dengue ^c	37.2	8.6	67.3	15.5
With climate	42.6	9.6	73.5	15.6
Support vector regression				
Correlation	54.6	15.1	90.5	18.0
PCMCi	54.6	15.3	90.4	18.8
Only dengue	55.3	14.9	90.5	17.7
With climate	53.0	14.8	91.1	19.9
Seasonal naive (baseline)	80.8	14.1	206.4	31.2

Abbreviations: MAE, mean absolute error; PCMCi, Peter-Clark momentary conditional independence; RMSE, root mean squared error.

^a Lowest median RMSE.

^b Lowest median MAE.

^c Lowest mean MAE and mean RMSE (up to 3 decimal places).

a single model for each city, random forests was chosen most frequently (Figure 1). This result held regardless of the error metric chosen as the minimizing criterion.

The models selected on the validation sets were trained on the full training data and evaluated on the holdout test set. The best overall model, the random forests algorithm trained only on dengue variables, predicted monthly dengue with a median MAE of 12.2 across all cities on unseen data (Table 4). Performance was improved slightly when selecting different combinations of algorithms and independent variables for each city individually, resulting in an MAE of 11.9 monthly cases for the median city, as well as lower mean RMSE and MAE. Both approaches outperformed the naive baseline forecast.

Qualitatively, the model was able to capture different kinds of time series, as illustrated in 3 sample cities in Figure 2. São Paulo experiences annual dengue outbreaks, though the sizes of the peaks vary across years. Sorocaba is a special case of this situation, where there was a single dominant outbreak in the 2015 statewide epidemic, with low monthly cases in the remaining years. Belém had a

more irregular time series of dengue cases, with multiple outbreaks per year and strong variation in the intensity of outbreaks. In all 3 cities, the models captured the qualitative changes well, both when selecting the best model for the specific city and when using the best model overall.

DISCUSSION

This study demonstrated the potential of machine learning models to forecast monthly dengue in Brazil. Our model outperformed existing comparable approaches by achieving a lower forecast error for São Paulo than the model of Baquero et al. (26).

Many of the public health measures available for curbing the spread of dengue target the vector—for example, insecticide spraying, environmental management, or education of the public on safe water storage (6). Vector surveillance can help focus these measures on infested areas. However, surveillance of infected mosquitoes is expensive—Brazil already spends over 1 billion BRL (\$190 million US) annually on vector control (48, 49)—and mosquito density is

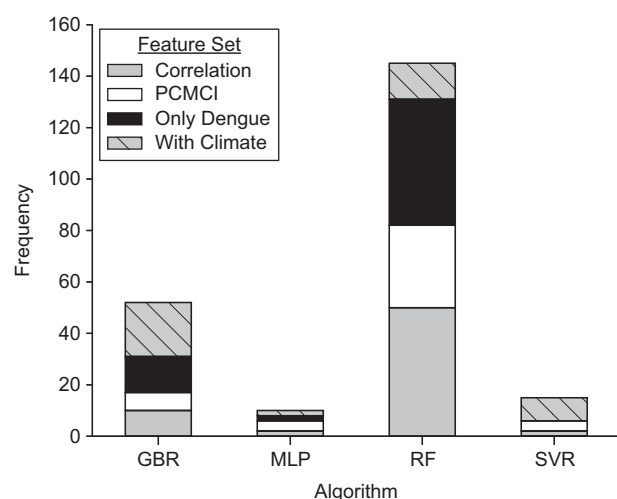


Figure 1. Distribution of the optimal combination of dengue prediction algorithm and input feature set for 234 cities (lowest mean absolute error), Brazil, 2007–2015. The bars show how often each algorithm (x-axis) and input feature set (color) is selected as the optimal combination for each Brazilian city to predict dengue incidence during the validation period (2007–2015). GBR, gradient boosting regression; MLP, multilayer perceptron; PCMCi, Peter-Clark momentary conditional independence; RF, random forests; SVR, support vector regression.

not always correlated with disease incidence (50). Monitoring human infections complements vector surveillance and may signal an impending outbreak. Improvements in early warning systems may contribute to evidence-based policies and an efficient use of public health resources—for example, for mosquito control activities (51, 52).

Brazil has strong epidemiologic surveillance infrastructure, obligatory dengue reporting, and a commitment to open data. Leveraging these data, the model developed in this study 1) is cost-effective, 2) adapts to policy needs, and 3) may integrate with existing surveillance systems:

1. The model developed in this study has a low computational cost after training and makes new predictions in few seconds. The model also requires few data points. On aggregate, epidemiologic data alone are not

only sufficient but optimal. No other data need to be collected, cleaned, or stored.

2. The model supports both regional and local policies. The general model may inform statewide or national planning; the city-specific model has optimal local performance while still leveraging data on dengue dynamics across the country. The model can also be adjusted to different prediction horizons, independent variables, or levels of spatiotemporal granularity. For example, in this study we used monthly epidemiologic data, since they were available for a longer time period, although weekly data were also available from 2014 onward.
3. Finally, the model can easily be integrated into the existing surveillance infrastructure. For example, several Brazilian cities use nowcasting for decision-making, which corrects for reporting delays by estimating the present number of dengue cases (53). Using the nowcasted data as inputs, our model may enhance surveillance efforts by providing forecasts of the expected outbreak trajectory based on state-of-the-art machine learning methodologies.

Some limitations of this study must also be considered. In 2015, Brazil recorded large outbreaks of dengue, especially in the state of São Paulo (54). Climate conditions and a historic water crisis in 2014–2016 were considered to be the primary causes of this spike in cases (55). Increased private water storage near residences created extra breeding habitats for mosquitoes, and extreme drought has been shown to increase the risk of dengue, especially in urban areas (56, 57). However, there is also evidence to suggest that some early cases of the 2015–2016 Zika epidemic were misdiagnosed and misreported as dengue, thus raising the dengue case count (58–61). The cocirculation of dengue and Zika viruses posed a challenge for diagnosis and surveillance. It is difficult to assess the impact misdiagnosis had on data quality in 2015, since reporting was not required until the World Health Organization proclaimed a public health emergency of global concern in February 2016 (62).

In Web Appendix 1 (available at <https://academic.oup.com/kwac090>), we consider the potential effect of this data bias on our predictions. We train 2 separate models on data from 2007–2015, with and without the year 2015, respectively. We note that the 2 models perform similarly well on

Table 4. Dengue Prediction Errors on the Test Data Set, Brazil, 2016–2019

Model	MAE		RMSE	
	Mean	Median	Mean	Median
Random forests (only dengue) ^a	53.7	12.2	130.4	25.4
City-specific model ^b	52.5	11.9	119.2	26.5
Seasonal naive (baseline)	134.7	20.0	293.9	42.6

Abbreviations: MAE, mean absolute error; RMSE, root mean squared error.

^a Lowest median RMSE.

^b Lowest mean MAE, median MAE, and mean RMSE.

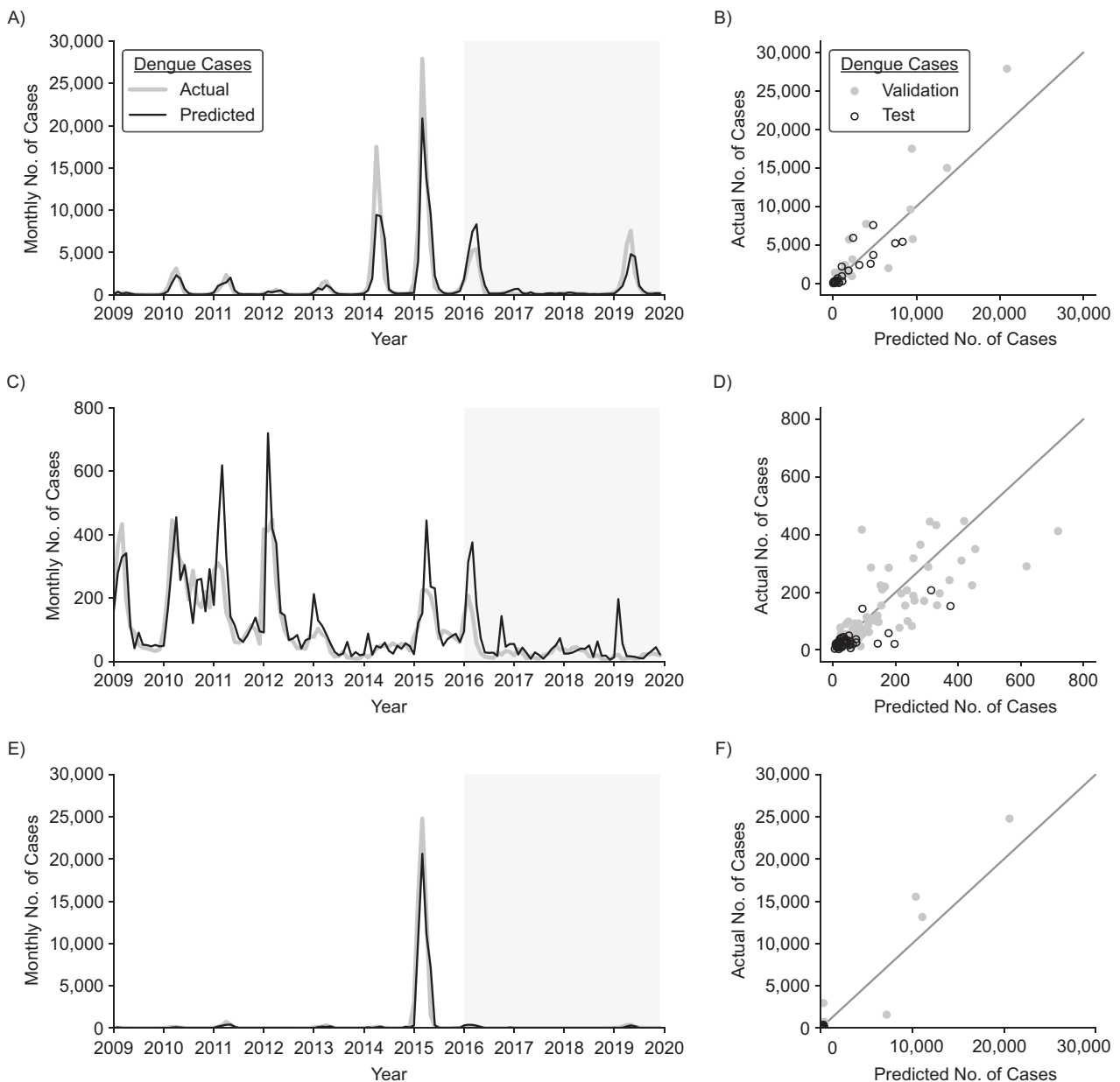


Figure 2. Predicted and actual numbers of dengue cases in São Paulo (top row), Belém (middle row), and Sorocaba (bottom row), Brazil, during 2007–2019. The left-hand column (panels A, C, and E) shows the time series of actual (gray) and predicted (black) numbers of cases in each city for both the validation (2009–2015) and test (2016–2019; shaded region) data sets. The right-hand column (panels B, D, and F) shows the predicted number of cases (x-axis) versus the actual number of cases (y-axis) for the validation (gray) and test (black-outlined) data sets.

the test data set; the difference in the errors is not significant ($P > 0.8$). The results from this sensitivity analysis suggest that inclusion of the 2015 data is not a serious concern for the predictive model (Web Table 1, Web Figure 1).

Machine learning is a valuable addition to the epidemiologist's toolbox (63). Yet, a drawback of the machine learning models used in this study relative to other epidemiologic approaches is low explainability. Effective disease prevention requires an understanding of its causes in addition to the expected incidence. We can assess the importance of differ-

ent predictors of dengue (e.g., Shapley additive explanations (SHAP) values), but this does not tell us which interventions could effectively reduce the disease burden.

A benefit of using causal discovery for feature selection is the potential to better understand the causes of dengue outbreaks. While causal inference has been a cornerstone of epidemiologic studies, the artificial intelligence community has only relatively recently incorporated causality—for example, to increase generalizability. A machine learning model that incorporates causality may transfer better to

changes in data collection or study locations (64). PCMCi and other causal discovery methods, however, rely on strong assumptions that may limit this benefit in the context of our study. Specifically, PCMCi requires 1) causal sufficiency, 2) the causal Markov condition, and 3) the faithfulness assumption (37). In this study, we did not include all variables shown to be linked to dengue, such as human mobility (7) or land use (8). This does not affect the PCMCi algorithm's ability to improve predictions, but it limits potential causal interpretations. In future work, investigators may expand the types of independent variables and causal discovery algorithms used to better understand potential causes.

Causal feature selection was not the optimal method across all cities in our study in terms of prediction accuracy, possibly given the strong autocorrelation in the dengue time series for the 1-month prediction horizon. This result contributes to our understanding of the role of causal feature selection for applications in epidemiology. Similar findings on benchmark data sets (35) demonstrate that causal feature selection does not always offer the most accurate predictions, but it may produce other benefits, such as reduced covariate shift. Similarly, any real-world implementation of the model presented in this paper must face a trade-off between the most accurate predictions and more robust, interpretable model inputs.

Because the impact of climate conditions on *Aedes* populations is well-documented, meteorological variables are often used for dengue forecasts (26, 28–30, 65, 66). The results of our analysis suggest that the importance of climate variables is not uniform across all cities in Brazil. This aligns with recent findings on the heterogeneity of predictors of dengue seasonality, intensity, and trend across the mesoregions of Brazil (54). It has also been shown that the most predictive time windows of variables such as temperature and rainfall vary across Brazilian state capitals (67). Brazilian cities are very heterogeneous, not only in terms of their climate zones but also given disparities in income and human development. Forecasts must adapt to regional differences. At the same time, predictive performance increases when including data from all cities across Brazil, rather than just training models on individual city time series. In Web Appendix 2, we begin to explore whether certain socioeconomic and geographic city characteristics can help us determine in advance whether predictions will benefit from climate variables. The results suggest that population size and human capital help determine whether climate variables are helpful for dengue forecasting in a given location (Web Figure 2, Web Table 2, Web Table 3), though further research is needed and other predictors should be explored.

In conclusion, we compared machine learning algorithms and sets of independent variables, and showed that a random forests model trained on 11 months of historical dengue cases was effective at forecasting dengue 1 month ahead in over 200 Brazilian cities. In some cities, errors on unseen data were reduced even further by adopting a city-specific approach and selecting one of the other machine learning models. We therefore find that there is no universal model and that appropriate predictors may vary across cities, confirming recent findings in other geographic contexts (68). The most appropriate approach will depend on the intended

use case, such as allocating public health resources across cities or developing an early warning system for outbreaks in a specific location.

ACKNOWLEDGMENTS

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This work was supported by grant 2019/26595-7 from the São Paulo Research Foundation.

All raw data sets used in this study are publicly available. Cleaned data and software code are available on GitHub (69).

Conflict of interest: none declared.

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